

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () before the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Derive Poisson's Equation. Also, write the Laplace equation.	CO1	6
B) Explain the charge simulation method for the evaluation of electric field distribution.	CO1	6
C) Explain stressed oil volume breakdown mechanisms in a liquid dielectric material.	CO2	6
Q.2 Solve Any Two of the following.		12
A) Explain the treeing and tracking phenomenon in solid dielectric materials.	CO2	6
B) What is Paschen's law? Derive the expression for the minimum breakdown voltage.	CO2	6
C) Explain the generation of high-frequency AC voltage with the help of the Tesla coil.	CO3	6
Q.3 Solve Any Two of the following.		12
A) Explain Townsend's breakdown criterion. Also, derive the current growth equation in the presence of the primary process.	CO2	6
B) Describe the working of a Van-de-Graff generator circuit with a neat sketch.	CO3	6
C) Write a short note on Intrinsic, electromechanical, and chemical breakdown in solid insulation systems.	CO2	6
Q.4 Solve Any Two of the following.		12
A) Explain the type and routine test performed on the power transformer.	CO1	6
B) Explain the reasons for overvoltage due to switching surges and methods to minimize it.	CO4	6
C) Explain the generation of impulse voltage using a multistage Marx circuit. Also, describe its important parts.	CO3	6

Q. 5	Solve Any Two of the following.		12
A)	Explain with a schematic the working of a cascade transformer to generate high AC voltage.	CO3	6
B)	Explain the lightning phenomenon by elaborating on the concept of the dart leader and the stepped leader.	CO4	6
C)	Explain various tests required to be performed on high-tension cables.	CO1	6

***** End *****